Project D: RQ1 and RQ2 Descriptive Analysis

Who drives AI-in-education research in Europe? Based on the OpenAlex candidate dataset (EU-affiliated papers, 2020 to 2026).

Data and method

The analysis uses 7,594 EU-affiliated candidate papers (ALL) collected from five OpenAlex queries (75,126 records collected, 67,684 unique after deduplication by OpenAlex ID). The provisional STRICT set contains 3,668 of these papers whose venue matched the Scopus source list by ISSN with coverage for the publication year. STRICT stays provisional until the team confirms the exact inclusion rules.

Institution counts show each paper once per institution. Country counts use full counting: a paper with authors in several countries counts for each, so the 27 country counts sum to 9,050, more than the 7,594 papers. Venue names exist only in the STRICT file, so venue results describe the STRICT set (3,668 records, 1,402 venues).

RQ1: Which European institutions and venues drive research since 2020?

Institutions. The University of Eastern Finland has the most papers (96), followed by CNRS (93), the University of Bonn (86), Leibniz University Hannover (80) and the Technical University of Munich (78). The University of Oulu (74), LMU Munich (71), University College Dublin (67), the University of Salamanca (66) and the Bucharest University of Economic Studies (64) complete the top ten. Germany holds six of the top fifteen places and Finland three. Output is highly skewed: of 2,489 institutions, the median has 2 papers and only 361 have 10 or more.

Countries. Germany (1,382 papers), Spain (1,137) and Italy (908) lead, followed by France (560), Greece (481), the Netherlands (473), Finland (418), Sweden (405), Portugal (365) and Romania (342). Germany, Spain and Italy together account for 3,427 country-paper counts, about 45% of the 7,594 papers, with overlap from co-authorship.

Venues. The 3,668 STRICT records spread over 1,402 venues, and the top ten cover only 20%. By records, the two largest venues are Springer proceedings book series: Communications in Computer and Information Science (155) and Lecture Notes in Networks and Systems (136). Among journals, the leaders are Education Sciences (84),

Frontiers in Education (75), Applied Sciences (68), IEEE Access (48) and Sustainability (47). Alzheimer's & Dementia appears with 43 records, which is almost certainly off-topic and points to noise from the broad search queries.

Sensitivity to relevance. Recomputing the institution ranking with only the 6,442 papers flagged "Likely relevant" keeps the University of Eastern Finland first (87), then Bonn (79), Hannover (75), TU Munich (69) and CNRS (68). CNRS falls from second to fifth, Salamanca leaves the top ten and L3S Research Center enters (63). The leading group is stable, but exact positions are not. The keyword screen is weak (it flags Alzheimer's & Dementia records as likely relevant), so manual validation is still needed.

RQ2: Macro-level trends and the effect of Scopus filtering

Yearly trend. ALL output rose from 329 papers in 2020 to 2,062 in 2025, more than sixfold.

Year	ALL	STRICT	STRICT share	ALL growth
2020	329	139	42.2%	n/a
2021	369	207	56.1%	+12.2%
2022	420	236	56.2%	+13.8%
2023	725	365	50.3%	+72.6%
2024	1,308	633	48.4%	+80.4%
2025	2,062	992	48.1%	+57.6%
2026*	2,381	1,096	46.0%	+15.5%

*2026 is a partial year, so its count and growth rate are not comparable with full years. It already exceeds 2025.

Growth was modest until 2022 and then accelerated, with the largest increases in 2023 (+73%) and 2024 (+80%). International bibliometric studies report the same pattern, with accelerating growth from 2023 following the release of generative AI tools such as ChatGPT. The timing is consistent with that explanation, but a descriptive count cannot show that those tools caused the rise.

ALL vs STRICT. STRICT keeps 3,668 of 7,594 records (48.3%). The yearly share was 56% in 2021 and 2022 and has fallen to 46% in 2026 (2020 starts lower, at 42%). STRICT follows the same growth shape as ALL, so the filter mainly changes volume, not the trend. A

possible reason for the falling share is that newer venues are not yet on the provisional Scopus list, but this analysis does not test that.

Limitations

- 2026 is an incomplete year.
- STRICT is provisional until the Scopus inclusion rules are confirmed.
- Relevance screening is keyword-based and the manual audit is incomplete; the 7,594 papers are candidates, not confirmed AI-in-education papers.
- Country counts overlap through co-authorship (full counting).
- Venue results cover only the STRICT subset.
- Institution counts use OpenAlex institution names and identifiers as given, so some variants of the same organisation may be split.